

National University of Computer and Emerging Sciences, Lahore Campus



Course:	Computer Organization and Assembly Language	Course Code:	EE213
Program:	BS(Data Science)	Semester:	Fall 2022
Duration:	50 Minutes	Total Marks:	50
Paper Date:	Oct, 2022	Weight:	
Section:		Page(s):	2
Exam:	Quiz-6		

Name: _____ Roll No. _____ Section: _____

Q1. Write the contents of "str2" after executing the following program.

[org 0x0100] [1x5 Marks]

```

mov si, str1
mov di, str2
mov cx, [len]
add di, cx
dec di
push ds
pop es
cld
next: movsb
sub di, 2
loop next
mov ax, 0x4c00
int 0x21
len: dw 17
str1: db 'Never odd or even'
str2: db 'Final Examination'
    
```

Solution:

never so odd even

Q2. How many times int 0x01 will come in the following code? [1x5 Marks]

pushf; setting TF=1

pop ax

or ax, 0x100

push ax

popf ✓

mov cx, 5 ✓

rep stosb ✓

mov ax, 0x4c00 ✓

int 21h ✓

7 times

Solution:

Q3. What is the final value of DX register (in hex) after executing the following code?

```

mov di, str2
push ds
pop es
cld
mov cx, [len]
xor dx, dx
again: jcxz done
repne cmpsb
jne done
inc dx
jmp again
done: mov ax, 0x4c00
int 0x21
len: dw 8
str1: db 'Ukraine.'
str2: db 'Romania.'

```

dm 0002

Solution:

Q4. The video display memory is divided into 25 rows and 80 columns. Assume that for the purpose of this question, rows are numbered from 0 to 24 and columns are numbered from 0 to 79. Assuming there are no errors in the code and the screen is cleared prior to execution of this code, write the output of the following code in the right column. With each output line, also write the row and the column number. For example, if the output 'A' is generated at row 10 and column 20, then write as follows: [1x15 Marks]
Row=10, Column=20, Output=A

```

[org 0x0100]
mov ax, 0xb800
mov es, ax
mov bl, 2
mov di, 0
mov dx, 0x0730
mov cx, 9
print: mov ax, cx
add dx, ax
div bl
cmp ah, 0
je skip
mov ah, 0
add di, 320
mov word[es:di], dx
skip: mov dx, 0x0730
dec cx
jnz print
mov ax, 0x4c00
int 0x21

```

Solution:

Row	Column	Output
2	0	9
4	0	7
6	0	5
8	0	3
10	0	1